

PART A
(Answer ALL questions)

Maximum Marks : 60

(10 × 2 = 20)

- I. ✓ (a) Lithium at 20°C is BCC structure and has a lattice constant of 0.35092 nm. Calculate the atomic radius of a lithium atom in nanometers.
- ✓ (b) In the solidification of a metal, what is the difference between an embryo and a nucleus? What is the critical radius of a solidifying particle?
- ✓ (c) Describe any two surface treatments of steel. Why are these treatments carried out?
- ✓ (d) Draw a continuous-cooling transformation diagram for a eutectoid plain-carbon steel.
- ✓ (e) What is the difference between the slip and twinning mechanisms of plastic deformation of metals?
- ✓ (f) What type of phase diagram is necessary for a binary alloy to be precipitation - hardenable. Comment on the process.
- (g) What are some of the typical characteristics of ceramic materials?
- (h) What are the four basic types of stainless steels?
- (i) Explain the applications of shape memory alloys.
- (j) Compare between recovery and recrystallisation process.

PART B

? (4 × 10 = 40)
direction.

- ✓ II. (a) Explain how the atomic density of crystallographic planes and (density) can be determined. What is the importance of highly packed planes and directions?
- (b) Distinguish between line and planar defects.

OR

- ✓ III. (a) Explain how the equi-axed and columnar grains can be formed during solidification process.
- (b) Explain the Fick's law of diffusion and its importance.

- ✓ IV. (a) Reconstruct the iron-carbon phase diagram indicating all key phases, reactions and microstructures.
- (b) Describe various forms of metallic solid solutions and explain their differences.

OR

- ✓ V. (a) Distinguish between martempering and austempering process.
- (b) Compare annealing with normalizing process.

VI. (a) Creep & Stress rupture difference
(b) Fatigue fracture characteristics

VII (a) strain hardening vs. grain boundary strengthening
(b) elastic vs. inelastic behavior

VIII (a) Cast iron - classification & application
(b) Titanium - importance in aerospace
(c) Copper - 5
(d) Aluminum - 2

(P.T.O.)